

Artificial Intelligence in Pediatric Radiology

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- Describe the big players in radiology AI, both general and pediatric, since 2023
- Summarize what radiology AI does well and what remains unresolved

Academic output. PubMed (E-utilities) yearly counts per query and per modality/task term group; **OpenAlex** citation counts (union of modality/task searches, deduped), ranked separately for 2008-2022 and 2023-present because citations favor older papers; **DBLP** for ML venues; **PatentsView** for granted patents.

Clinical problems. The pediatric radiology-AI query AND a term group per problem (bone age, fracture, pneumonia, appendicitis, brain tumor, ...), counted per era; a paper can name several problems.

Datasets. Public pediatric imaging datasets, sizes verified against the primary papers or hosting pages.

Commercial players. The **FDA AI-enabled device list** (public spreadsheet: decision date, device, company, lead panel), filtered to the Radiology panel; cross-checked against a curated list of products with pediatric indications.

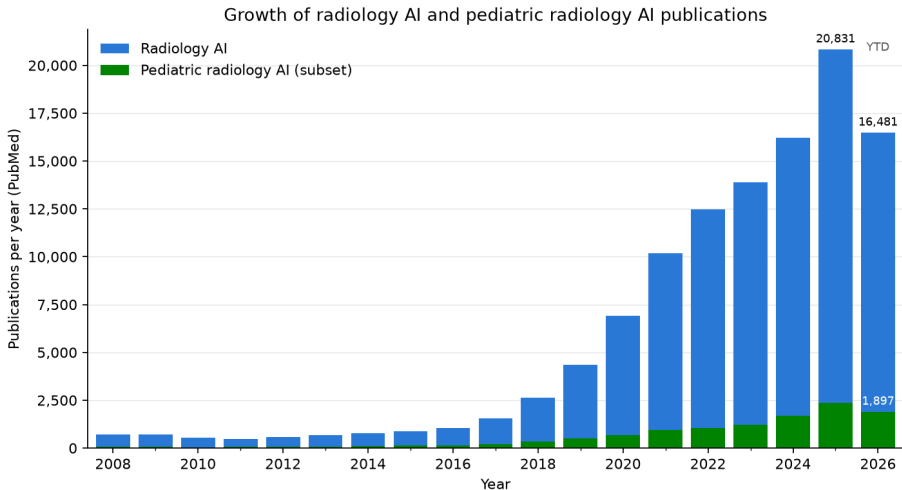
Trade press. Newsletter archives (The Imaging Wire, RSNA News, TLDR, Signify Research, Radiology Business) split into stories; a story is radiology-AI when AI terms co-occur with imaging terms, pediatric when pediatric terms also co-occur.

Pediatric filter. PubMed: the radiology-AI query AND (pediatric* OR paediatric* OR child* OR infant* OR neonat* OR adolescen* OR "children's hospital"). Most-cited lists: the title must also carry a pediatric term (bone age, fetal, newborn, ...), otherwise highly cited adult papers float in. Newsletters: pediatric term inside the same story.

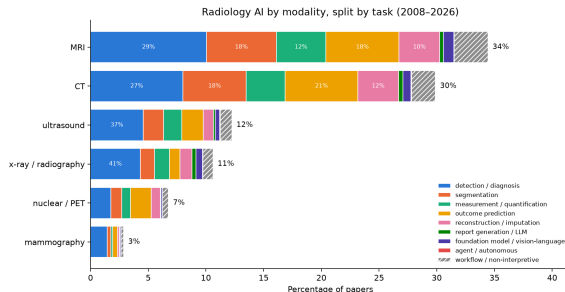
Representative PubMed query (radiology $\cap$ AI):

(radiology OR radiograph* OR "medical imaging" OR MRI OR "computed tomography" OR CT[tiab] OR ultrasound[tiab] ... NOT ("optical coherence" OR fundus OR dental OR histopatholog* ...)) AND ("artificial intelligence" OR "deep learning" OR "convolutional neural network" OR radiomics ...)

Radiology AI publication counts



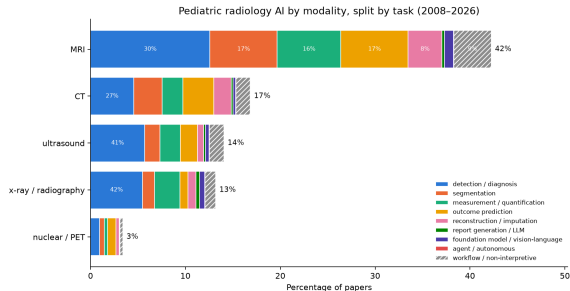
Radiology AI publications, stratified by modality and task



Bar = percentage of radiology-AI papers (98,842, 2008–present) whose title/abstract names the modality; segments = which task terms those papers use (overlapping). What each task means, by what the model produces:

- **detection / diagnosis:** is a finding present now, and which one (label or box): fracture, pneumonia, tumor; includes triage and screening
- **segmentation:** outline a structure or lesion (a mask)
- **measurement / quantification:** a number from the image: bone age, Cobb angle, organ volume, fetal biometry
- **outcome prediction:** a future risk, response, or survival estimate from the image (prognosis, radiomics signatures)
- **reconstruction / imputation:** a better or missing image: lower dose, faster scans, denoising, synthetic CT/MR
- **report generation / LLM:** text: draft, summarize, or extract from radiology reports
- **foundation model / vision-language:** a large pretrained model reused across tasks (segment-anything, self-supervised)
- **agent / autonomous:** multi-step actions taken without a human in the loop
- **workflow / non-interpretive:** protocoling, scheduling, ordering, decision support, education

Pediatric radiology AI publications, stratified by modality and task



Same construction, restricted to the pediatric subset (10,141 papers). Top pediatric modalities: MRI 43%, CT 17%, ultrasound 14%, x-ray / radiography 13%. Mammography is omitted: its 76 pediatric-query hits (of 10,141) are adult breast-imaging papers whose abstracts mention children.

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Cross-check: an independent 2025 scoping review of pediatric radiology AI

Kamran et al., *Pediatric Radiology* 2025 (doi 10.1007/s00247-025-06462-5): 789 original pediatric-focused articles hand-screened from four databases, 2005 to August 2024. Our PubMed counts are larger because they include any paper that mentions a pediatric term.

- **Modality:** radiography 38%, MRI 33%, ultrasound 14%, CT 11%, nuclear 2%. Our pediatric query ranks MRI first (MRI 43%, CT 17%, ultrasound 14%, x-ray / radiography 13%) because MRI-heavy neurodevelopment papers mention children.
- **Subspecialty:** musculoskeletal 33%, neuro 29%, chest 17%, body 14%, cardiac 6%.
- **Top applications:** bone age (88 articles), pneumonia (65), brain tumors (54), scoliosis (45), hip dysplasia (28), congenital heart disease (25), autism (18), epilepsy (14), ADHD (9), hydrocephalus (9).
- **Use of AI:** 91% image interpretation / diagnosis; 5.6% image quality; under 2% for acquisition, communication, protocoling, education, and policy combined.
- **Data:** 65% used a single local hospital dataset; 7% did not report the dataset; China 28%, USA 25%, Canada 7% of articles.
- **Growth:** 23 articles in 2018, 158 in 2022, 171 in 2023, consistent with our PubMed curve.
- **Recommendations:** multi-institutional, age-diverse pediatric datasets; CLAIM-style reporting with subgroup results and calibration; funding for the neglected non-interpretive areas (policy, education, implementation); children and families as stakeholders across the AI lifecycle.
- **What it adds to this deck:** the same three leading tasks (bone age, pneumonia, brain tumor) and the same gap (small local datasets, adult tools applied to children) reached by a manual review.

The biggest public pediatric radiology AI datasets

Dataset	Year	Modality	Size	Ages	Access
CBTN clinical MRI (Children's Brain Tumor Network)	2023	brain / spine tumor MRI	23,101 exams (1,526 patients); tumor masks for 370	pediatric	application (NCI CCDI)
GRAZPEDWRI-DX	2022	wrist radiographs (trauma)	20,327 images (6,091 patients)	0.2–19 y	open
RSNA Pediatric Bone Age Challenge	2017	hand radiographs	14,236 images (bone age + sex)	mean 10.6 y	open (RSNA terms)
FETAL_PLANES_DB	2020	fetal ultrasound (standard planes)	12,400 images (1,792 pregnancies)	fetal	open
ABCD Study	2018–	brain MRI, longitudinal	11,878 participants, up to 7 follow-ups	9–10 y at baseline	application (NBDC)
PediCXR / VinDr-PCXR	2023	chest radiographs	9,125 studies; 36 findings, 15 diagnoses	< 10 y	credentialed (PhysioNet)
Guangzhou pediatric chest X-ray (Kermayn)	2018	chest radiographs	5,856 images (normal / bacterial / viral pneumonia)	1–5 y	open
ABIDE I + II (autism)	2012 / 2016	brain MRI (structural, resting fMRI)	2,156 subjects	5–64 y (mostly children)	open
Regensburg Pediatric Appendicitis	2023	abdominal ultrasound + labs	1,709 images (579 patients)	0–18 y	open
dHCP (Developing Human Connectome Project)	2022–24	neonatal + fetal brain MRI	886 neonatal + 297 fetal datasets, with segmentations	20–45 weeks	open (NDA)
Pediatric-CT-SEG (TCIA)	2022	chest / abdomen / pelvis CT	718 series (359 patients); 29 organ contours	5 days – 16 y	open
BraTS-PEDs (pediatric brain tumor challenge)	2023–25	brain tumor MRI	464 patients (2024 challenge); 1,649 series on TCIA	pediatric	registration / TCIA
FeTA (fetal brain tissue annotation)	2021–24	fetal brain MRI	280 volumes (120 train + 160 test); 7 tissue labels	18–35 weeks	registration

Sizes verified against the primary paper or hosting page (2026-09). Only two pediatric radiograph sets exceed 10,000 images; the largest public pediatric-only CT set has 359 patients; no RSNA pediatric challenge has run since bone age in 2017. Adult benchmarks (CheXpert 224k, MIMIC-CXR 377k) exclude children.

Most-cited radiology AI papers, 2008-2022

Cites	Paper	What it answers
15,130	A survey on deep learning in medical image analysis (Medical Image Analysis, 2017)	<i>survey</i> : What can deep learning do across medical imaging? (the field's reference survey)
9,402	nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation (Nature Methods, 2020)	<i>segmentation</i> : Can one self-configuring method segment any organ or lesion without hand tuning? (nnU-Net)
6,720	Computational Radiomics System to Decode the Radiographic Phenotype (Cancer Research, 2017)	<i>radiomics</i> : Can standardized image features (radiomics) be extracted reproducibly? (PyRadiomics)
4,921	Deep Learning in Medical Image Analysis (Annual Review of Biomedical Engin. . . , 2017)	<i>survey</i> : How does deep learning apply to medical image analysis? (review)
4,873	Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning (Cell, 2018)	<i>classification</i> : Can transfer learning diagnose pediatric pneumonia on chest X-ray and retinal disease on OCT?
4,702	Convolutional neural networks: an overview and application in radiology (Insights into Imaging, 2018)	<i>tutorial</i> : What is a convolutional neural network, explained for radiologists?
3,768	Artificial intelligence in radiology (Nature reviews. Cancer, 2018)	<i>review</i> : How will AI change radiology practice? (Hosny et al.)
3,187	COVID-Net: a tailored deep convolutional neural network design for detection of COVID-19 cases from ches. . . (Scientific Reports, 2020)	<i>classification</i> : Can a CNN detect COVID-19 on chest X-ray? (COVID-Net)
2,690	CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison (AAAI Publications (The Associatio. . . , 2019)	<i>dataset</i> : A 224k chest-radiograph dataset with uncertainty labels for training and benchmarking (CheXpert)
2,432	Covid-19: automatic detection from X-ray images utilizing transfer learning with convolutional neural ne. . . (Physical and Engineering Sciences. . . , 2020)	<i>classification</i> : Can transfer learning detect COVID-19 on X-ray with small data?

Most-cited radiology AI papers, 2023-present

Cites	Paper	What it answers
1,435	TotalSegmentator: Robust Segmentation of 104 Anatomic Structures in CT Images (Radiology Artificial Intelligence, 2023)	<i>segmentation</i> : Can one open tool segment 100+ anatomic structures on any CT? (TotalSegmentator)
776	Redefining Radiology: A Review of Artificial Intelligence Integration in Medical Imaging (Diagnostics, 2023)	<i>survey / review</i>
737	Diffusion models in medical imaging: A comprehensive survey (Medical Image Analysis, 2023)	<i>survey / review</i>
607	MRI-based brain tumor detection using convolutional deep learning methods and chosen machine learning te... (BMC Medical Informatics and Decis... , 2023)	<i>oncology</i>
567	A Study of CNN and Transfer Learning in Medical Imaging: Advantages, Challenges, Future Scope (Sustainability, 2023)	<i>other</i>
564	How Artificial Intelligence Is Shaping Medical Imaging Technology: A Survey of Innovations and Applicati... (Bioengineering, 2023)	<i>survey / review</i>
492	Medical image analysis using deep learning algorithms (Frontiers in Public Health, 2023)	<i>other</i>
483	Brain Tumor Detection Based on Deep Learning Approaches and Magnetic Resonance Imaging (Cancers, 2023)	<i>oncology</i>
454	DCSAU-Net: A deeper and more compact split-attention U-Net for medical image segmentation (Computers in Biology and Medicine, 2023)	<i>segmentation</i>
445	Medical image data augmentation: techniques, comparisons and interpretations (Artificial Intelligence Review, 2023)	<i>other</i>

Recent citations accrue to surveys first; the clinical landmarks of this era (TotalSegmentator, pancreatic cancer detection on non-contrast CT, Sybil lung-cancer risk) sit among many review articles.

Most-cited pediatric radiology AI papers, 2008-2022

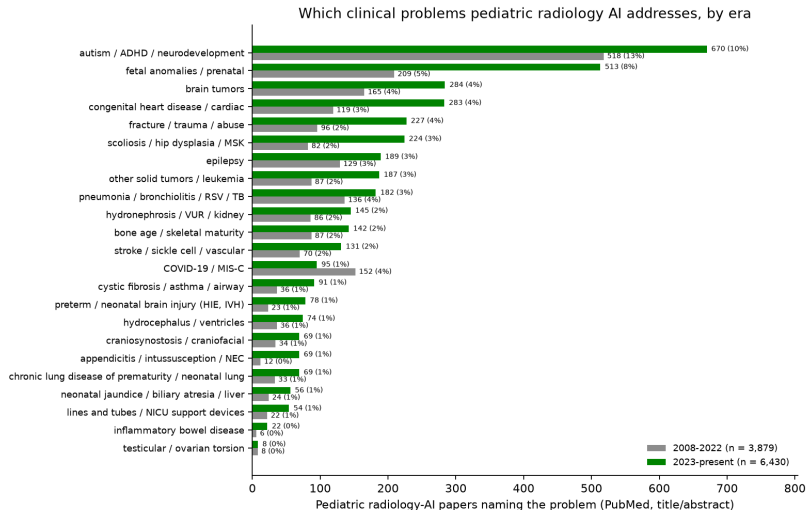
Cites	Paper	What it answers
473	The RSNA Pediatric Bone Age Machine Learning Challenge (Radiology, 2018)	<i>bone age</i> : How well can crowdsourced AI estimate bone age? (RSNA challenge, 260 teams)
469	Performance of a Deep-Learning Neural Network Model in Assessing Skeletal Maturity on Pediatric Hand Rad... (Radiology, 2017)	<i>bone age</i> : Can a CNN assess skeletal maturity as accurately as radiologists?
454	Fully Automated Deep Learning System for Bone Age Assessment (Journal of Imaging Informatics in... , 2017)	<i>bone age</i> : Fully automated bone age from hand radiographs
435	Deep learning for automated skeletal bone age assessment in X-ray images (Medical Image Analysis, 2016)	<i>bone age</i> : Deep learning for automated bone age (BoNet)
204	Improving Image Quality and Reducing Radiation Dose for Pediatric CT by Using Deep Learning Reconstructi... (Radiology, 2020)	<i>reconstruction</i> : Can deep-learning reconstruction cut pediatric CT dose while improving image quality?
166	Computerized Bone Age Estimation Using Deep Learning Based Program: Evaluation of the Accuracy and Effic... (American Journal of Roentgenology, 2017)	<i>bone age</i> : Is a commercial deep-learning bone-age program accurate and faster in practice?
134	Regression Convolutional Neural Network for Automated Pediatric Bone Age Assessment From Hand Radiograph (IEEE Journal of Biomedical and He... , 2018)	<i>bone age</i> : Regression CNN for pediatric bone age
126	Artificial intelligence-assisted interpretation of bone age radiographs improves accuracy and decreases... (Skeletal Radiology, 2018)	<i>bone age</i> : Does AI assistance make radiologists' bone-age reads more accurate and consistent?
116	Deep Learning Based Attenuation Correction of PET/MRI in Pediatric Brain Tumor Patients: Evaluation in a... (Frontiers in Neuroscience, 2019)	<i>reconstruction</i> : Can deep learning replace CT for PET/MRI attenuation correction in children?
108	Using a Dual-Input Convolutional Neural Network for Automated Detection of Pediatric Supracondylar Fract... (Investigative Radiology, 2019)	<i>fracture</i> : Can a CNN detect supracondylar fractures on pediatric elbow radiographs?

Most-cited pediatric radiology AI papers, 2023-present

Cites	Paper	What it answers
62	Fetal brain tissue annotation and segmentation challenge results (Medical Image Analysis, 2023)	<i>fetal MRI</i> : How well do algorithms segment fetal brain tissue? (FeTA challenge)
50	Automated tumor segmentation and brain tissue extraction from multiparametric MRI of pediatric brain tum... (Neuro-Oncology Advances, 2023)	<i>segmentation</i>
47	Review on deep learning fetal brain segmentation from Magnetic Resonance images (Artificial Intelligence in Medici... , 2023)	<i>segmentation</i>
46	Deeplasia: deep learning for bone age assessment validated on skeletal dysplasias (Pediatric Radiology, 2023)	<i>bone age</i>
39	Learning to segment fetal brain tissue from noisy annotations (Medical Image Analysis, 2023)	<i>segmentation</i>
38	MRI-Based End-To-End Pediatric Low-Grade Glioma Segmentation and Classification (Canadian Association of Radiologi... , 2023)	<i>segmentation</i>
37	A Dempster-Shafer Approach to Trustworthy AI With Application to Fetal Brain MRI Segmentation (IEEE Transactions on Pattern Anal... , 2024)	<i>segmentation</i>
33	Deep Learning-Based Multiclass Brain Tissue Segmentation in Fetal MRIs (Sensors, 2023)	<i>segmentation</i>
31	Bone Age Assessment Using Artificial Intelligence in Korean Pediatric Population: A Comparison of Deep-L... (Korean Journal of Radiology, 2023)	<i>bone age</i>
29	Automatic biometry of fetal brain MRIs using deep and machine learning techniques (Scientific Reports, 2023)	<i>fetal MRI</i>

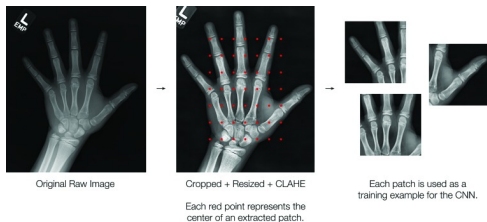
Fetal brain MRI segmentation, pediatric brain tumor segmentation, and bone age lead the recent era; citation counts are two orders of magnitude below the adult list.

Which medical problems pediatric radiology AI addresses



The RSNA Pediatric Bone Age Machine Learning Challenge

Halabi et al., Radiology 2019 (2008–2022 era, most-cited pediatric paper)



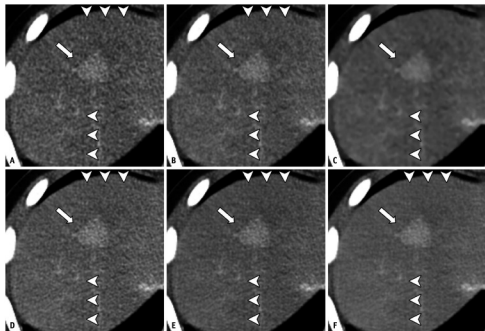
Halabi et al., Radiology 2019, figure 4: hand radiograph,
cropped and contrast-normalized, with the patches one team
used

source: cdn.ncbi.nlm.nih.gov

- **Problem:** bone age from a left-hand radiograph: a tedious, subjective read that every children's hospital does daily.
- **Data:** 14,236 hand radiographs from two children's hospitals (Stanford, Colorado) with Greulich–Pyle bone age by pediatric radiologists; 200 test cases read by six reviewers.
- **Method:** an open competition (260 teams); entries were convolutional networks that regress bone age from the image plus sex.
- **Result:** the winning model's mean absolute difference from the reviewers was 4.2 months; 48 teams were within 5 months, i.e. at reviewer-to-reviewer variability.
- **Why it matters:** the first public pediatric imaging benchmark; it made bone age the one pediatric task with many models and cleared products.

Deep-learning reconstruction for pediatric CT: image quality at lower dose

Nagayama et al., Korean J Radiol 2022 (open-access companion to Brady et al., Radiology 2021, the most-cited paper on this topic)



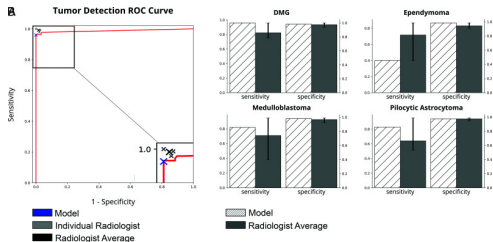
Nagayama et al., Korean J Radiol 2022, figure 6: the same pediatric abdominal CT reconstructed six ways (FBP, hybrid iterative, deep learning)

source: cdn.ncbi.nlm.nih.gov

- **Problem:** pediatric CT dose is limited by noise; the standard fixes (filtered back projection, hybrid iterative reconstruction) trade noise for a blotchy texture and lost detail.
- **Data:** 120 pediatric abdominopelvic CT scans (median age 5 years) reconstructed with FBP, two iterative settings, and three deep-learning-reconstruction strengths.
- **Method:** the vendor's deep-learning reconstruction (a network trained to map low-dose to high-dose raw data); noise, texture, edge sharpness, and reader scores compared across reconstructions.
- **Result:** deep-learning reconstruction cut noise the most while keeping texture and sharpness, giving the highest reader scores; Brady et al. reported the same in 2021 with a 52% dose reduction at equal or better image quality.
- **Why it matters:** this is the pediatric AI benefit available on today's scanners: lower dose or shorter scans without an interpretive model in the loop.

Deep learning for pediatric posterior fossa tumor detection and classification

Quon et al., AJNR 2020, multi-institutional



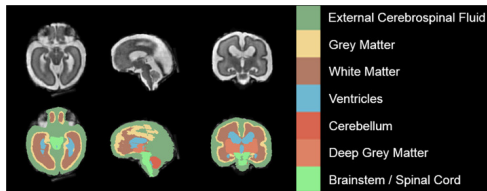
Quon et al., AJNR 2020, figure 1: T2 MRI examples of the four posterior fossa tumor types

source: cdn.ncbi.nlm.nih.gov

- **Problem:** the four common posterior fossa tumors of childhood (medulloblastoma, pilocytic astrocytoma, ependymoma, diffuse midline glioma) need different surgery and prognosis, and preoperative MRI reads disagree.
- **Data:** 617 children from five institutions: T2-weighted MRI of 199 tumors and 418 controls.
- **Method:** a 2D convolutional network (ResNeXt-50) trained on axial T2 slices to detect a tumor and assign its type; performance compared with four radiologists.
- **Result:** tumor detection accuracy 92% (AUC 0.99); four-class tumor typing accuracy 72%, similar to the radiologists, whose own accuracy ranged widely.
- **Why it matters:** a multi-site pediatric neuro-oncology model that matched human readers; the next step, prospective use, has not been shown.

FeTA: the fetal brain tissue annotation and segmentation challenge

Payette et al., Sci Data 2021 (dataset) and Med Image Anal 2023 (challenge results; most-cited pediatric paper of 2023–present)



Payette et al., Sci Data 2021, figure 2: fetal brain MRI with
the seven tissue labels

source: cdn.ncbi.nlm.nih.gov

- **Problem:** fetal brain MRI volumes of cortex, white matter, ventricles, and cerebellum track abnormal development, but manual segmentation takes hours per case and moves with the fetus.
- **Data:** 80 (later 120) reconstructed fetal brain MRI volumes, 20–35 weeks, normal and pathological (spina bifida), each with seven tissue labels.
- **Method:** a MICCAI challenge: 21 teams submitted segmentation networks (most nnU-Net variants) evaluated on held-out cases.
- **Result:** the best entries reached a mean Dice of about 0.79 across tissues; performance dropped on the youngest and most abnormal brains.
- **Why it matters:** the first shared benchmark for fetal MRI; a template for the multi-center datasets pediatric radiology lacks.

Deeplasia: deep learning for bone age assessment validated on skeletal dysplasias

Rassmann et al., *Pediatr Radiol* 2023

Deeplasia: deep learning for bone age assessment validated on skeletal dysplasias

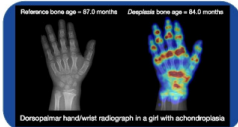
- Development and validation of a method for assessing the bone age in children with skeletal dysplasias
- A prior-free deep-learning approach with an ensemble of three convolutional neural networks

Deeplasia is demonstrated to:

- achieve a competitive mean absolute difference of 3.87 months on the test dataset of the Radiological Society of North America
- generalize to cases with molecularly-confirmed diagnosis of achondroplasia, hypochondroplasia, pseudohypoparathyroidism, Noonan, Silver-Russell and Ullrich-Turner syndromes, and a mutation in the SHOX gene

**Pediatric
Radiology**

Rassmann S. et al., 2023



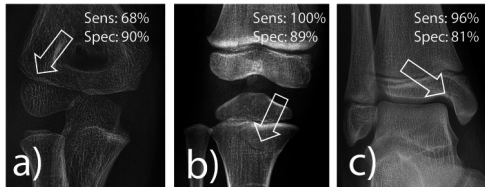
- **Problem:** bone-age models are trained on ordinary children, but the patients who most need the read (achondroplasia, Turner, Noonan, Silver–Russell, SHOX) have abnormal hands.
- **Data:** the RSNA challenge set for training; testing on the RSNA test set, the Digital Hand Atlas, and 568 radiographs from children with molecularly confirmed dysplasias.
- **Method:** an ensemble of three convolutional networks with no hand-crafted priors, released as open source.
- **Result:** mean absolute difference 3.9 months on the RSNA test set and about 5–6 months on dysplasia cohorts, comparable to inter-rater variation.
- **Why it matters:** shows how to test a pediatric model on the rare phenotypes a children's hospital actually sees, not only on the benchmark.

Rassmann et al., *Pediatr Radiol* 2023, graphical abstract

source: cdn.ncbi.nlm.nih.gov

Real-life benefit of AI fracture detection in a pediatric emergency department

Ziegner et al., Eur Radiol 2025



Ziegner et al., Eur Radiol 2025, figure 4: pediatric fractures

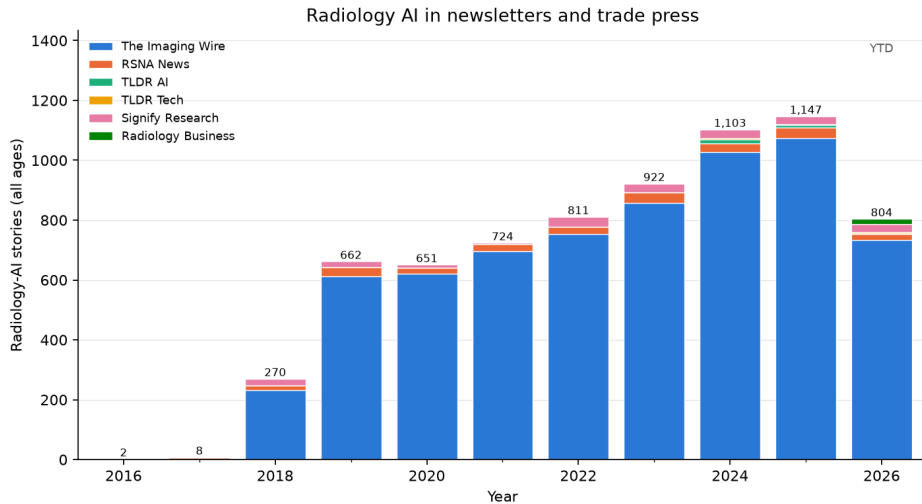
the software found, with its per-region sensitivity and

specificity

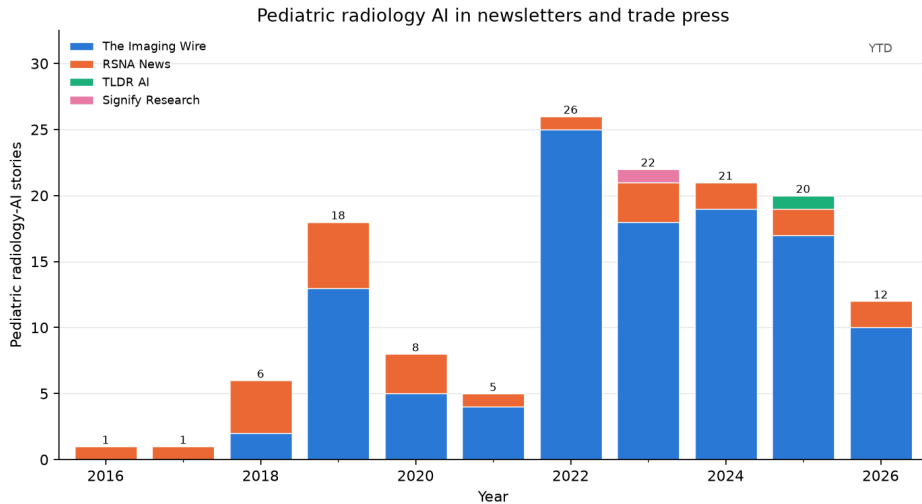
source: cdn.ncbi.nlm.nih.gov

- **Problem:** pediatric fractures are subtle and often first read by inexperienced physicians; cleared fracture tools were validated on adults.
- **Data:** 1,672 real-life radiographic studies (1,657 trauma patients, upper and lower extremity) from one children's emergency department.
- **Method:** a commercial fracture-detection tool (pediatric indication) run stand-alone, then three inexperienced readers with and without the AI output.
- **Result:** stand-alone sensitivity about 90% with region-dependent specificity; readers with AI gained only a few percentage points of accuracy (about 2% of wrong reads corrected, 4% of correct reads reversed).
- **Why it matters:** a good stand-alone model translated into a limited reader benefit: the gap between retrospective accuracy and clinical value that most pediatric AI has not yet crossed.

Newsletters — radiology AI stories, all ages



Newsletters — pediatric radiology AI stories



Pediatric radiology AI in the trade press, 2023 (22 stories)

Date	Newsletter	Story
2023-01-08	The Imaging Wire	Gestational Age AI
2023-01-25	The Imaging Wire	ChatGPT Authoring Research
2023-03-01	The Imaging Wire	Fetal Ultrasound AI Generalizability
2023-03-17	The Imaging Wire	When Sao Paolo's Diagnosticos da America (DASA, the world's fourth largest diagn
2023-04-07	The Imaging Wire	Ultrasound Spots Breech Pregnancies
2023-04-07	The Imaging Wire	When Sao Paolo's Diagnosticos da America SA (DASA, the world's fourth largest di
2023-04-12	Signify Research	Where do opportunities exist?
2023-04-12	RSNA News	RSNA Webinar to Examine Impact of Large Language Models
2023-05-07	The Imaging Wire	New C-Arm Launched
2023-06-07	The Imaging Wire	BrightHeart's Fetal Heart AI
2023-06-12	RSNA News	Understanding the Imaging Spectrum of non-Hodgkin Lymphoma in Children, Adolescents and Young Adults
2023-06-19	The Imaging Wire	Better Together at SIIM
2023-06-25	The Imaging Wire	AI Detects Child Abuse on CT
2023-07-07	RSNA News	Survivors of Childhood Cancer: Risk of Future Disease, Screening Needs and the Role of Radiology
2023-08-02	The Imaging Wire	MRI Radiomics Hones in on Crohn's
2023-08-02	The Imaging Wire	Sonio Gets FDA Nod for Fetal AI
2023-09-20	The Imaging Wire	GE Lands \$44M AI Ultrasound Grant
2023-10-08	The Imaging Wire	ChatGPT Reports for Patients
2023-11-08	The Imaging Wire	Philips to Expand AI-Powered POCUS
2023-11-08	The Imaging Wire	Stories of Success in Cloud and AI
2023-11-12	The Imaging Wire	Uneven Success Against Breast Cancer Pediatric Radiation
2023-11-26	The Imaging Wire	AI of MRI Diagnoses Autism

Pediatric radiology AI in the trade press, 2024 (21 stories)

Date	Newsletter	Story
2024-01-10	The Imaging Wire	AI Models Go Head-to-Head in Project AIR Study
2024-01-17	The Imaging Wire	Echo AI's RHD Impact
2024-01-20	The Imaging Wire	Us2.ai
2024-02-07	The Imaging Wire	Meet Gleamer at ECR 2024
2024-03-20	The Imaging Wire	AI Helps Residents Detect Fractures
2024-04-22	The Imaging Wire	GE Launches US Scanners
2024-05-19	The Imaging Wire	Early Preeclampsia Risk Screening
2024-06-29	The Imaging Wire	FDA Clears New Clarius Prenatal App
2024-07-10	RSNA News	Experts Outline Considerations to Deploy AI in Radiology
2024-08-14	The Imaging Wire	Cardiac Ultrasound Scanner Cleared
2024-08-25	The Imaging Wire	When Follow-Up Falls Short Pediatric CT Use
2024-08-25	The Imaging Wire	MAUI Emerges from Stealth
2024-09-04	The Imaging Wire	Samsung Finalizes Sonio Acquisition
2024-09-08	The Imaging Wire	AZmed AI Extended to Kids
2024-10-06	The Imaging Wire	Fracture AI Improves Over Time
2024-10-10	The Imaging Wire	Low-Dose CT in Kids Hinton Nabs Nobel, Cancels MRI
2024-10-10	The Imaging Wire	Low-Dose CT Confounds CAD in Kids
2024-11-19	RSNA News	Incorrect AI Advice Influences Diagnostic Decisions
2024-11-25	The Imaging Wire	Fetal Ultrasound AI Cleared
2024-12-02	The Imaging Wire	Milvue Joins deepc AI Platform
2024-12-15	The Imaging Wire	Researchers Study AI for Pediatric Fractures

Pediatric radiology AI in the trade press, 2025 (20 stories)

Date	Newsletter	Story
2025-01-08	The Imaging Wire	AI for Pediatric Fracture Detection
2025-01-08	The Imaging Wire	Children's National taps Microsoft to prototype AI tools .
2025-01-29	The Imaging Wire	Samsung Launches OB/GYN Ultrasound
2025-02-02	The Imaging Wire	Ultrasound AI Detects Fetal Defects
2025-02-27	The Imaging Wire	AI Detects Pediatric Epilepsy
2025-03-04	TLDR AI	AI to diagnose invisible brain abnormalities in children with epilepsy (4 minute read)
2025-04-02	The Imaging Wire	AZmed Gets New Chest AI Clearances
2025-04-13	The Imaging Wire	GE's Cincinnati Pediatric Partnership
2025-05-14	The Imaging Wire	Philips Partners with NVIDIA on AI for MRI
2025-05-18	The Imaging Wire	New GE Clearances
2025-05-20	RSNA News	Radiologists Share Tips to Prevent AI Bias
2025-06-01	The Imaging Wire	BrightHeart's 3rd FDA Clearance
2025-07-03	The Imaging Wire	DeepEcho Gets FDA Nod for Fetal AI
2025-07-10	RSNA News	How Is AI Being Used in Daily Neuroradiology Practice?
2025-08-03	The Imaging Wire	Radiologist Pay Jumps Nearly 8% in New Survey
2025-08-13	The Imaging Wire	AI-Assisted Fracture Detection
2025-08-27	The Imaging Wire	Sonio Launches New Maternal-Fetal AI Options
2025-10-01	The Imaging Wire	Carebot Expands AI Portfolio
2025-10-08	The Imaging Wire	Reducing CT Radiation Dose System-Wide
2025-11-09	The Imaging Wire	AI Detects Pediatric Elbow Fractures

Pediatric radiology AI in the trade press, 2026 (year to date) (12 stories)

Date	Newsletter	Story
2026-01-07	The Imaging Wire	BrightHeart Raises EUR 11M
2026-01-28	The Imaging Wire	AHA Publishes New Stroke Guidelines
2026-02-01	The Imaging Wire	DL-Based CT Reduces Radiation Dose
2026-02-08	The Imaging Wire	GE Bolsters Ultrasound with Fetal AI
2026-03-29	The Imaging Wire	Ultrasound AI Gets Breakthrough Nod
2026-04-05	The Imaging Wire	Imaging Growth Slows
2026-05-10	The Imaging Wire	Pediatric MRI Safety, RTs and Remote Radiology, and PSMA-PET
2026-06-22	RSNA News	Researchers Use AI to Establish Body Charts for CT Imaging Across the Adult Lifespan
2026-06-28	The Imaging Wire	BrightHeart's Bright Future
2026-07-12	The Imaging Wire	AI's Rise in Pediatric Imaging
2026-08-16	The Imaging Wire	CT vs. MRI for Pediatric TBI, Helium Vulnerability, and Ditching the Disk
2026-08-25	RSNA News	AI Challenges Fuel Innovation Beyond the Leaderboard

Commercial software with a pediatric angle

Vendor	Product	Task	Pediatric status	FDA list
Gleamer	BoneView	fracture detection on radiographs	pediatric indication (age 2+) cleared 2023	4 (2022–2025)
AZmed	Rayvolve	fracture (and chest) findings on radiographs	pediatric fracture indication cleared 2024	4 (2022–2025)
Visiana	BoneXpert	automated bone age (GP / TW)	pediatric by design; CE-marked since 2009 (not on the FDA AI list)	not listed
16 Bit	Rho	automated bone age	pediatric by design	1 (2024)
Ever Fortune.AI	EFAI Bonesuite Bone Age Pro	automated bone age	pediatric by design	11 (2022–2026)
BrightHeart	Fetal EchoScan	fetal echocardiography view / anomaly assist	prenatal (fetal) by design	5 (2024–2025)
GE HealthCare / Canon / Siemens / Philips	TrueFidelity, AiCE, Deep Re- solve, Precise Image	deep-learning CT / MR reconstruction (lower dose, faster scans)	adult-cleared scanner software, widely used for pediatric dose reduction	307 (2014–2026)
Subtle Medical / AIRS Medical	SubtleMR, SwiftMR	accelerated MRI via deep-learning de-noising	adult-cleared; pediatric scan-time studies	13 (2018–2026)
Qure.ai	qXR / qER	chest radiograph and head CT triage / quantification	adult indications; pediatric TB-screening evidence only	9 (2020–2026)
Aidoc / Viz.ai	BriefCase, Viz LVO / ICH	worklist triage (hemorrhage, LVO, PE, pneumothorax)	adult-only indications; the largest deployed category	43 (2018–2026)

“FDA list” = number of radiology AI-enabled devices the company has on the FDA list (years). Pediatric status is the publicly stated indication; confirm the age range in the 510(k) summary before purchase.

Worth knowing as a radiologist — and why

- **TotalSegmentator** (open tool): free, one-command segmentation of 100+ structures on any CT (and MR); the basis for opportunistic measurements (bone density, muscle, organ volumes)
- **nnU-Net** (open tool): if a vendor claims a segmentation result, this is the baseline they had to beat
- **MONAI** (open framework): what your data scientists will build with; knowing the name lets you read a methods section
- **RSNA Pediatric Bone Age Challenge (2017)** (paper / dataset): the one pediatric task with a public benchmark, many models, and cleared products (BoneXpert, Rho)
- **"Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs" (Zech et al., PLoS Med 2018)** (paper): the cleanest demonstration that a model can learn the hospital instead of the disease; the reason to demand local validation
- **CLAIM checklist (Radiology: AI 2020)** (guideline): how to tell a well-reported AI study from a weak one
- **"Improving Image Quality and Reducing Radiation Dose for Pediatric CT by Using Deep Learning Reconstruction" (Brady et al., Radiology 2021)** (paper): the most-cited non-bone-age pediatric paper: deep-learning reconstruction lowers pediatric CT dose with better image quality; the most immediately usable pediatric benefit
- **FDA AI-enabled device list** (registry): the authoritative list of what is cleared; check the indication's age range before buying
- **Gleamer BoneView / AZmed Rayvolve** (products): the first fracture-detection tools with pediatric indications; realistic first pediatric AI purchases
- **MedSAM / segment-anything models** (foundation models): where annotation and segmentation tooling is heading: click, don't draw

- **Finding what is there.** Detection / diagnosis is the bulk of the literature (47% of radiology-AI papers, 46% of pediatric); in adults it has become deployed worklist triage, the largest block of the 1,164 FDA-listed radiology AI devices.
- **Outlining and measuring.** Segmentation (24% of papers) has open, strong defaults (nnU-Net, TotalSegmentator); in children the mature measurement task is bone age (RSNA challenge: 4.2-month error, several cleared products), and measurement / quantification is 22% of pediatric papers.
- **Better images from less dose.** Reconstruction / imputation is 15% of radiology AI and 10% of pediatric work; deep-learning reconstruction is on today's scanners and cut pediatric CT dose by about half at equal or better image quality.
- **Shared benchmarks where they exist.** Fetal and neonatal brain MRI (FeTA, dHCP) and pediatric brain tumors (BraTS-PEDs) now have challenge datasets, and multi-site pediatric neuro-oncology models match radiologists retrospectively.

- **Foundation and vision-language models:** 4.5% of radiology-AI papers (3.5% pediatric); segment-anything style tools (MedSAM) and generalist radiology models are replacing task-specific training.
- **Report generation and LLMs:** 3.1% of papers; drafting and extracting from reports, with fluency ahead of reliability.
- **Agents:** 0.5% of radiology-AI papers (33 pediatric papers in total); LLMs that call imaging tools and take multi-step actions.
- **Pediatric-specific:** bone age robust to skeletal dysplasias (Deeplasia); fetal ultrasound AI is the most active commercial pediatric area in the trade press (BrightHeart, Sonio, DeepEcho clearances, 2023–2026); adult fracture tools extended to children (Gleamer 2023, AZmed 2024) and now tested in a real pediatric emergency department.
- **Generalization studies:** adult-trained CT organ segmentation degrades in small children, and the first pediatric prospective-style reader studies show smaller gains than stand-alone accuracy implies.

Open problems — where a children's hospital could contribute

- **Which diseases?** Since 2023 the pediatric corpus (6,430 papers) leans to autism / ADHD / neurodevelopment (670); fetal anomalies / prenatal (513); brain tumors (284); congenital heart disease / cardiac (283). Barely studied: testicular / ovarian torsion (8); inflammatory bowel disease (22); lines and tubes / NICU support devices (54); neonatal jaundice / biliary atresia / liver (56); craniosynostosis / craniofacial (69); appendicitis / intussusception / NEC (69). Which of these deserve a model first is an open question for this group.
- **Dataset curation.** What would a multi-center pediatric dataset look like: age-stratified, consented, protocol-diverse, with rare phenotypes? Which of our own archives (CT, ultrasound, NICU radiographs) could become the pediatric benchmark that does not yet exist?
- **Validation.** Local, age-stratified testing of adult-cleared tools; prospective reader studies rather than retrospective accuracy; monitoring for drift as children grow and protocols change.
- **Beyond interpretation.** Workflow / non-interpretive work is 13% of papers, but communication with families, education, protocoling and policy are almost absent; these are cheaper to build and evaluate than diagnostic models.
- **Regulation and liability.** Few of the 1,164 radiology clearances carry a pediatric indication; which adult clearances are acceptable to use off-label in children, and under what local validation?

- ① **Buy maturity, build for the gaps:** adopt cleared adult-derived tools that transfer (reconstruction, bone age, fracture with a pediatric indication); treat the under-studied pediatric problems as local validation and research.
- ② **Demand local pediatric validation** before clinical use, by age group, and check the age range in the clearance.
- ③ **Prioritize dose and throughput:** deep-learning reconstruction gives the clearest pediatric benefit today.
- ④ **Invest in data:** a curated, shareable pediatric dataset is the scarcest resource in the field and the contribution a children's hospital is uniquely placed to make.